

Nina Grant

Ph.D. Candidate, Rutgers University

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/in/nina-grant-2021



EDUCATION

- Ph.D. Rutgers University, Atmospheric Science** 2021-2026 (exp)
Focus on climate intervention impacts, GPA: 4.0
Advisors: Alan Robock & Lili Xia
- M.S. Rutgers University, Atmospheric Science** 2023
Focus on agricultural impacts of climate intervention, GPA: 4.0
Advisors: Alan Robock & Lili Xia
- B.A. Princeton University, Geosciences** 2021
Focus on climate science and sustainability
Advisor: Gabriel Vecchi

RESEARCH EXPERIENCE

- Research Assistant** Aug 2021-present
Rutgers Impact Studies of Climate Intervention Lab New Brunswick, NJ
Advisors: Dr. Alan Robock and Lili Xia
Tropical Agriculture Responses to Stratospheric Aerosol Intervention
- Investigated how stratospheric aerosol injections (SAI) impact Indian wheat and rice crop suitability and production using CESM2 ARISE-SAI-1.5 model outputs
 - Researching the impact of climate intervention and nuclear war on coffee and cocoa yields in Brazil and Ghana using multi-model downscaled ISIMIP data
 - Comparing the skill and suitability of machine learning and statistical downscaling methods for SAI scenarios
- Weather & Climate Data Analyst Summer Student Employee** Jun 2023 – Jan 2024
Electric Power Research Institute Remote
Tech Lead: Erik Smith
- Quantified the skill of 4 gridded datasets (ERA5, ERA5-Land, MERRA-2, and PRISM) at representing extreme temperatures and precipitation over the conterminous U.S.
 - Processed 40+ years of high-resolution climate data using Python (xarray, Dask, Rasterio) to evaluate dataset biases across temporal and spatial scales.
- Senior Thesis** Aug 2020- May 2021
Princeton University Princeton, NJ
Advisor: Dr. Gabriel Vecchi
Solar Geoengineering Effects on the Indian Monsoon Precipitation Patterns

- Analyzed GFDL climate model simulations of hemispherically varying volcanic eruptions and a 1% decline in solar radiation to isolate the impact of injection location and amount on monsoonal behavior
- Investigated whether local thermodynamics or large-scale dynamics were responsible for the pattern shifts

Internship

Dec 2020- May 2021

Princeton Climate Action Plan Emission Reduction Strategies Team

Princeton, NJ

Project Lead: Nic Choquette-Levy

- Researched decarbonization transitions for Princeton landscaping services by deducing current GHG emissions and sound pollution to provide recommendations to the municipality of Princeton
- Served as translator during landscaper interviews and translated documents to Spanish to disseminate team findings to local communities

Junior Independent Work

Sep 2019- Jan 2020

Princeton University

Princeton, NJ

Advisor: Dr. Gabriel Vecchi

Sea Surface Temperature Variability as A Predictor of Indian Monsoon Activity

- Revisited past work by Prof. Vecchi to verify whether, with more data, a predictive relationship persisted between breaks in the monsoon and sea surface temperatures in the Indian Ocean and the Bay of Bengal

Research Internship

Jun 2019 – Aug 2019

ALMA Observatory

Santiago, Chile

Advisors: Chentao Yang and Violette Impellizzeri

- Applied a new imaging technique for carbon tracing in deep-space, red-shifted galaxies acquiring experience with Linux OS, virtual machines, data analysis, presenting scientific research, and bilingual communication skills

PROFESSIONAL TRAINING

Young Voices of Science

Feb 2026 – Present

Hubbard Brook Research Foundation (Online)

- Eight-week training program led by expert science communication consultants designed to equip undergraduate and graduate students in environmental fields with effective methods for sharing their scientific work and passions with non-technical audiences.
- Participants also complete at least one outreach project or activity, putting what they learn into practice.

Leadership & No-Blame Problem Solving Certificate

Jun 2025

Rutgers University (Online)

- Asynchronous 10-module course on civic leadership and practical governance skills using the No-Blame Problem Solving method. Taught how average citizens can enact

local change. Covered citizen rights, local power structures, media literacy, and strategies for engaging in government decision-making.

AGU Local Science Partner Spring Congressional Visit Workshop Mar 2025
Washington, D.C.

- Full-day workshop on science policy engagement, science communication, media advocacy, and congressional visit preparation. Included training on op-ed writing, messaging strategy, and mock meetings with legislators.

Climate Change AI Virtual Summer School Aug 2024
Online

- 3-month course on the applications of AI and machine learning for tackling climate change problems with lectures and hands-on coding tutorials. Topics spanned agriculture, biodiversity, climate science, GHG accounting and monitoring, power systems, transportation, and more.

SciComp Workshop - Collaborative Coding with GitHub Dec 2023
AGU Fall Meeting 2023, San Francisco, CA

- 4-hr workshop on version control, git, and GitHub best practices and how to use GitHub for scientific research projects

Machine Learning Specialization Sep 2023
Online

- 10-week course hosted on Coursera by Stanford University and DeepLearning.AI on supervised learning (linear regression, logistic regression, neural networks, decision trees), unsupervised learning (clustering, anomaly detection), recommender systems, and reinforcement learning

CESM Tutorial Jul 2023
Boulder, Colorado

- Selective week-long intensive on the science and advancements behind the CESM climate model and practical training exercises on running custom model simulations

CESM AGU Workshop Dec 2022
AGU Fall Meeting 2022, Chicago, IL

- 4-hr workshop on running simple custom models in CESM on the AWS cloud

LEADERSHIP (VOLUNTEER/SERVICE)

Solar Geo Society Ambassador 2024-Present

AGU Local Science Partner 2024-Present

Organizer of the 1st Rutgers Atmospheric Science Grad Student Retreat 2024

Student Convener of AGU'24 Session GC009	2024
Co-Chair of the 2026 Gordon Research Seminar	2024-Present
Conference Coordinator of the Princeton Energy Association Fall Conference	2020
Reviewer: Earth's Future (2025)	

HONORS AND AWARDS

• Young Voices of Science Fellow	2026
• Sigma Xi Nomination	2025
• Atmospheric Science Graduate Student Travel Grant (\$2000)	2025
• Solar Geo Society Ambassador	2024
• Atmospheric Science Graduate Student Travel Grant (\$1200)	2024
• Elected Co-Chair of the 2026 Gordon Research Seminar	2024
• Atmospheric Science Graduate Student Travel Grant (\$1500)	2023
• RCEI Travel Award (\$500)	2023
• RCI Fund Travel Award (\$250)	2022
• AGU Student Travel Grant (\$1000)	2022
• Dean's Fellowship, Rutgers University School of Graduate Studies (\$49,736)	2021

PUBLICATIONS

[Google Scholar](#) | [ORCID: 0000-0003-4188-9275](#)

Grant, N., Kiniry, J., & Aziz, F. (2025) Modeling the impacts of climate change on cocoa. In *Theobroma cacao - Past, Present and Future Insights*. IntechOpen.

<https://doi.org/10.5772/intechopen.1011776>

Smith, E., **Grant, N.**, Luo, X. et al. (2025). Evaluating the ability of gridded climate datasets to capture temperature and precipitation trends and extremes. *Sci Rep*, 15(1), 12607.

<https://doi.org/10.1038/s41598-025-97570-7>

Grant, N., Robock, A., Xia, L., Singh, J., and Clark, B. (2025). Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future*, 13(1), e2024EF005262. <https://doi.org/10.1029/2024EF005262>.

Singh, J., Sahany, S., Xia, L., Robock, A., and **Grant, N.** Comparing quantile mapping and other statistical methods in downscaling rainfall for agriculture impacts. (Submitted)

Grant, N., Robock, A., Xia, L., Kiniry, J., and Clark, B. Can Solar Climate Intervention Save Coffee and Chocolate from Climate Change? (in prep.)

INVITED TALKS

Invited Talk, <i>A brief introduction to the physical science of SRM</i> , New York, NY	Sep 2025
Invited Talk, <i>The Modern Challenges of Geoengineering</i> , Global Affairs Canada, Virtual	Jul 2025
Invited Discussion Leader, <i>Climate Engineering: Processes, Uncertainties, Responses, and Impacts</i> , Gordon Research Conference on Climate Engineering, Lucca (Barga), Lucca, Italy	Feb 2024
Invited Talk, <i>How I Got Involved in Sustainability Work</i> , The Hewitt School, Virtual	May 2021

CONFERENCE PRESENTATIONS

Oral Presentation, <i>Keeping Coffee and Chocolate on the Table in a Warming World</i> , AGU Fall Meeting, New Orleans, LA, USA	Dec 2025
Poster Presentation, <i>Comparing Machine Learning and Traditional Downscaling Methods for Climate Projections Under Stratospheric Aerosol Intervention</i> , AGU Fall Meeting, New Orleans, LA, USA	Dec 2025
Oral Presentation, <i>Can stratospheric aerosol intervention save coffee and chocolate from climate change?</i> Degrees Global Forum, Cape Town, South Africa	May 2025
Poster Presentation, <i>Impacts on Indian Agriculture Due to Solar Climate Intervention Using Agroclimatic Indices</i> , Degrees Global Forum, Cape Town, South Africa	May 2025
Poster Presentation, <i>Can stratospheric aerosol intervention save coffee and chocolate from climate change?</i> AGU Fall Meeting, Washington, D.C., USA	Dec 2024
Poster Presentation, <i>Enhancing Climate Forcing Data for Crop Models: Addressing Challenges in Downscaling and Bias Correction under Climate Intervention and Nuclear Winter Scenarios</i> , AGU Fall Meeting, Washington, D.C., USA	Dec 2024
Poster Presentation, <i>Can Solar Climate Intervention Save Coffee and Chocolate from Climate Change?</i> 14 th GeoMIP Meeting, Ithaca, NY, USA	Jul 2024

Poster Presentation, <i>Impacts on Indian Agriculture Due to Solar Climate Intervention Using Agroclimatic Indices</i> , Gordon Research Conference on Climate Engineering, Lucca (Barga), Lucca, Italy	Feb 2024
Poster Presentation, <i>Impacts on Indian Agriculture Due to Solar Climate Intervention Using Crop Suitability and Agroclimatic Indices</i> , AGU Fall Meeting, San Francisco, CA, USA	Dec 2023
Poster Presentation, <i>Assessing the Ability of Reanalysis Data to Identify Local Extremes</i> , AGU Fall Meeting, San Francisco, CA, USA	Dec 2023
Poster Presentation, <i>Indian Agricultural Impacts under GeoMIP G6 Experiments</i> , AGU Fall Meeting, Chicago, IL, USA	Dec 2022
Poster Presentation, <i>Indian Monsoon Precipitation Changes in CESM2-WACCM GeoMIP6 Experiments</i> , Gordon Research Conference on Climate Engineering, Newry, ME, USA	Jun 2022

TEACHING EXPERIENCE

Guest lecturer – Climate Modeling Rutgers University (Remote) 16:107:544 Modeling of Climatic Change (Graduate)	Fall 2024
Led unit on a Python-based global climate model, deployed and tested model on Rutgers' Amarel HPC, created reproducible setup via GitHub, and developed assignments, tutorials, and exercises. Gave guest lecture, supported students through the assignment, and coordinated with the model developer.	
Tutor - Math Private (Remote)	Feb-Jul 2024
Tutored a high school student in algebra and geometry, raising GPA by 1.6 points	
Tutor - Math Rutgers School of Environmental and Biological Sciences	Sep 2022 – Apr 2023
Tutored 14 undergraduate students in college-level mathematics (geometry, algebra, calculus) in weekly group and individual sessions, developed lesson plans, coordinated schedules for virtual and in-person sessions, adapted to the needs of the students each session	

PROFESSIONAL AFFILIATIONS

Sigma Xi	2025-Present
National Association of Black Geoscientists (NABG)	2025-Present

American Association for the Advancement of Science (AAAS)	2024-Present
American Geophysical Union (AGU)	2022-Present
National Society of Black Engineers (NSBE)	2018-2021

SKILLS AND TOOLS

Programming languages: Python, Bash script, MATLAB, SQL, HTML

Python Packages: Xarray, Numpy, Pandas, Matplotlib, Cartopy, Salem, Seaborn, PyTorch, Scikit-learn, Keras, SciPy, PyAutoGUI

Applications and Platforms: VSCode, GitHub, Spyder IDE, MobaXTerm, DaVinci Resolve, HPCs (Cheyenne/Derecho, Amarel, Frontera)

Languages: English (native), Spanish (near native)